

Milestone 2 (17th April, 5pm)

Live Prototype: <https://com-480-data-visualization.github.io/les-vizionnaires/>

1. Project Goal

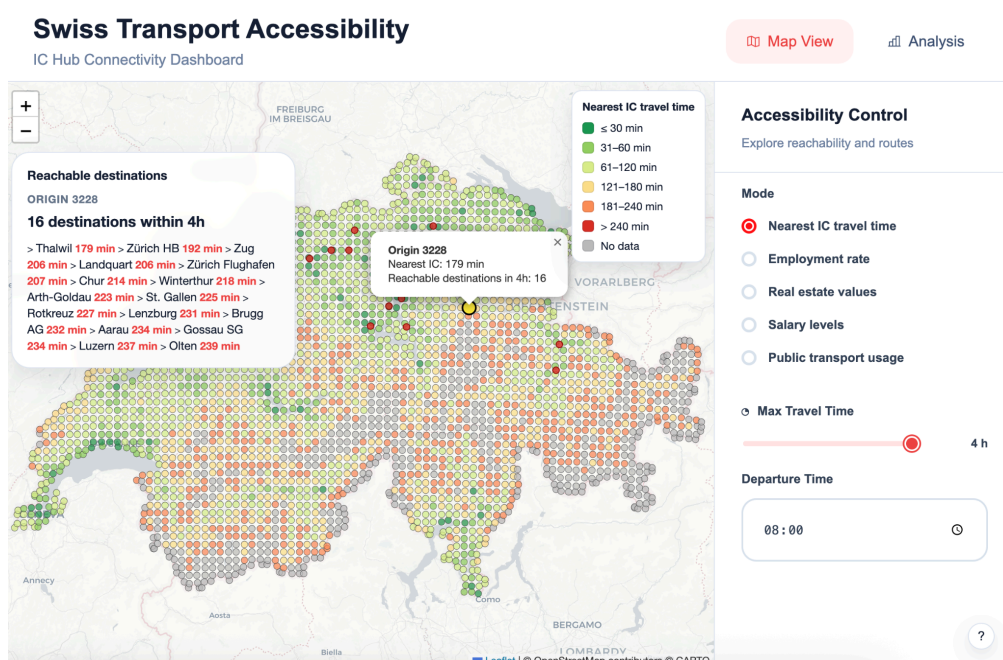
The primary objective of this project is to visualize and analyze the spatial accessibility of the Swiss public transport network, specifically focusing on the time required to reach major InterCity (IC) railway hubs. While Switzerland is renowned for its dense transport network, significant regional disparities exist.

By layering transport accessibility with socioeconomic indicators (real estate prices, employment density, average salary and passenger frequency), we aim to provide an interactive tool that reveals how connectivity correlates with urban development and territorial inequalities. Users should be able to identify "undervalued" regions (good accessibility but lower prices) or "isolated" hubs through an intuitive map-based interface and statistical analysis.

2. Visual Design & Sketches

Map View (The Spatial Dimension)

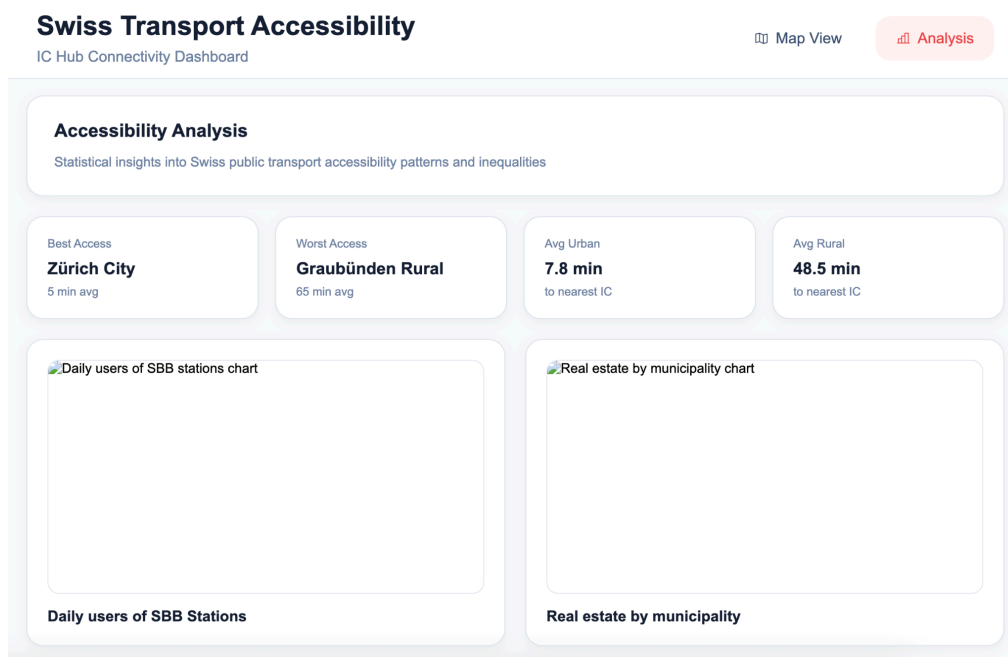
The main interface features a full-screen interactive map of Switzerland. We have opted for a grid-based encoding where each point represents a 4km² area.



- **Visual Encoding:** For the "Nearest IC travel time" mode, we use a divergent color scale (Green → Yellow → Red). Points that cannot reach a hub within the `MAX_TIME` threshold are rendered in Grey.
- **Layers:** Users can toggle between four thematic maps:
 - Travel Time to IC:** Proximity to major hubs.
 - Real Estate:** Average price per m².
 - Employment:** Concentration of jobs.
 - Salary:** Average salary per m².
 - Adoption Rate:** SBB station daily usage.
- **Interactions:** A sidebar includes sliders to adjust precomputed time-steps (e.g., 08:00, 12:00, 17:00, 22:00) and the maximum travel time threshold (30min to 4h). Clicking a point triggers a Detail-on-Demand popup.

Analysis View (The Statistical Dimension)

This view moves away from geography to focus on abstract data relationships using D3.js charts.



- **Correlation Scatter Plot:** X-axis (Travel Time) vs Y-axis (Real Estate Price) to identify outliers.
- **Bar Charts:** Comparison of accessibility metrics across different Cantons.
- **Distribution Histograms:** Percentage of the population living within specific time brackets of major infrastructure.

3. Tools and Lectures

Technical Stack

- **Data Processing:** `Python` with `r5py` to compute multimodal matrices using GTFS and OSM data.
- **Frontend:** `JavaScript`, `HTML5/CSS3`.
- **Map Engine:** `Leaflet.js` for handling the tiled map and point-grid overlays.
- **Visualization:** `D3.js` for the Analysis view charts and interactive widgets.

Relevant Lectures

- **Lecture 4 (Data & Data-driven documents):** We apply the Data-Join pattern (enter/update/exit) to link socioeconomic attributes to our grid. We use Scales to map quantitative data (travel time, prices) from the data domain to visual ranges (colors and positions).
- **Lecture 5 (Interaction & Interactive D3):** Following Shneiderman's mantra, we provide a "Map Overview" before allowing users to filter by time or zoom for "Details-on-demand." The "Analysis View" implements Coordinated Views to show the same data through different lenses (spatial vs. abstract).
- **Lecture 6 (Perception, color & Marks and channels):** Our grid uses Points as marks. We prioritize Position (the most accurate channel for quantitative data) for the coordinates, and Color Hue/Saturation to encode accessibility and prices, respecting Gestalt principles for grouping regional patterns.
- **Lecture 7 (Design for data viz & Do's and don'ts):** We follow the 4-level model to ensure our visual idioms match the data abstraction. We maintain Graphic Integrity by ensuring our color scales are perceptually uniform and our axes (in the Analysis View) start at zero to avoid the "Lie Factor."
- **Lecture 8 (Maps):** We address the challenges of Geographic Projections by using Leaflet.js. Our visualization combines a Grid-based map (for travel times) and thematic overlays, using GeoJSON logic for spatial data handling.
- **Lecture 11 (Tabular Data):** For the Analysis View, we will use Scatter Plots and Bar Charts to explore multidimensional relationships between accessibility, real estate, salary and employment, moving beyond simple spatial representation.

4. Project Breakdown

Core Visualization (MVP)

- [x] **Data Pipeline:** Extraction of IC Hubs from GTFS and grid generation.
- [x] **Precomputation:** Initial travel time matrix for a Friday at 08:00 AM.
- [x] **Functional Map:** Interactive Leaflet map with point-grid rendering.

- [x] **Basic UI:** Sidebar and tab navigation.
- [x] **Details-on-Demand:** Tooltips/Popups showing hub connectivity.

Extra Features (Milestone 3)

- **Multi-temporal Data:** Precomputing slots for 12:00, 17:00, and 22:00.
- **Socioeconomic Layers:** Integration of Real Estate, Employment, Salary and Adoption datasets as switchable map layers.
- **Advanced Analysis View:** Full D3.js suite with linked highlighting between charts and the map.
- **Visual Refinement:** Polishing the UI and potentially transitioning to hexagonal binning for better aesthetic density.

5. Prototype Review

The current prototype is fully operational and hosted on GitHub Pages. It features a robust navigation system between the Map View and Analysis View.

The core engine for the "Nearest IC travel time" is already implemented: the user can explore the Swiss territory, zoom in for regional details, and interact with the data points to get precise travel metrics. The precomputation logic in Python ensures a smooth user experience despite the complexity of the underlying multimodal routing. The current focus for the final milestone is the data integration for the three additional thematic maps and the development of the D3.js correlation charts to bridge the gap between mobility and socioeconomic factors.

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